ST 4: Destillation

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Group B.21
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1 Introduction

In this experiment, distillation using Raschig rings will be conducted on a mixture containing 10% mol ethanol and water. The goal of the experiment is to determine the time required for the column to reach steady-state conditions, and to evaluate the column's efficency as a function of vapour velocity. The purpose of the experiment is to provide insights to the fundamental principles of distillation, to understand how to achieve optimal operating conditions, and understand the key parameters influencing the operation of a distillation column.

2 Theory

2.1 Destillation

Destillation is a method used to purify or separate liquids from eachother. [1] The separation is based on the components having different boiling points, the mixed liquids is heated up so one of the components vaporizes and is separated from the liquid.

A packed column is a type of vertical column used for distillation and other separation processes. It contains one or more sections of packing material that increases the surface area for contact between phases, such as liquid and vapor. Fluid flows downwards and vapor rises upwards through the column, this promotes repeated vapor-liquid contact. [2]

2.1.1 Reflux

Reflux is a process used to to improve the separation of components in a mixture. It involves returning a portion of the cooled down vapor to the top of the distillation column to flow back down through the packing or trays. This helps the temperature of the reaction stay constant. This experiment operates with total reflux, witch means that all vapor is returned to the column. [3]

2.1.2 Column Efficiency

The column efficiency in a distillation column is defined as the ratio of the number of ideal equilibrium stages (theoretical plates) to the number of actual trays in the column. If the outgoing liquid and vapor phases reach equilibrium on each tray, then the distillation process is considered ideal. High column efficiency implies good contact between the gas and liquid phases, resulting in effective separation.

Since the separation process in a packed column does not occur in discrete trays or stages, the Height Equivalent to a Theoretical Plate (HETP), shown i Eq. (2.1), is used to estimate the equivalent number of theoretical stages (or "actual trays"). The HETP represents the height of packing material that achieves the same separation as one theoretical plate in a tray column. [4]

$$\text{HETP} = \frac{\text{packed height}}{\text{number of equivalent equilibrium stages}} \text{ [2]} \tag{2.1}$$

2.1.3 Flooding Point

Flooding is defined as a state where the liquid no longer can flow downward due to an overload of vapour going up. This results in the liquid being carried out of the column with the vapor or the liquid building up in the column. Therefore there is a maximum vapor flow rate for every given liquid flow rate, this is the flooding point. [2]

2.1.4 McCabe-Thiele Method

The McCabe-Thiele method is a graphic method used to determine the number of quilibrium stages, reflux ratio, and optimal feed stage location in a binary destillation. It's based on the

constants molar overflow assumption, that implies constant molar vapor and liquid flow rate leaving all stages in a section, and that the pressure is isobaric. It combines the equilibrium curve with operating lines to design and analyze distillation columns and helps visualize the distillation process and calculate necessary column specifications. [4]

2.2 Gas Chromatography

Gas chromatography is a method used to separate and analyze organic compounds. A injected gas phase is transported by a inert carrier gas (mobile phase) trough a column with a stationary phase. [5] The stationary phase is a liquid with a high boiling point or a solid adsorbent. As the sample moves through the column, different compounds interact differently with the stationary phase and thus separate based on their volatility and affinity for the stationary phase. After separation, the compounds exit the column and enter a detector. [6]

3 Experimental

The distillation experiment was a two part lab. The first part of the lab is to determine how long the column took to reach steady state condition, and the second part was to determine the efficiency of the column.

3.1 Apparatus

The Rasching-ring column with total reflux was used for the experiments. The the pump that controlled the reflux was connected to a computer.

3.2 Configurations

Firstly to configure the column the computer was turned on. Then the cooling water aswell. Next the sampling taps on the top and bottom of the column were closed. After these steps the column was filled with a mixture of 1596.1 mL of etanol and 4203.9 mL of water, so that the volume fraction of ethanol (x_{EtOH}) in the mixture was 0.10. Then the program on the computer was started by pressing start. The temperature was set to 105 °C and after the top temperature had changed significantly the power of the reboiler was set to 40%. The timer started when the vapor started to condense.

In case of flooding conditions, switch off the heater immediately. In any emergency situation, turn off the heater but leave the cooling water on.

3.3 Exercise nr. 1

For the first exercise the purpose was to determine how long the column took to reach steady state conditions.

When the timer was started, after the first condensation, there were drawn samples from the top tap every 5 minutes for an hour, and with the last sample there were drawn one sample from the bottom sampling tap. This totaled in 13 samples. The temperature and pumpflow was also noted for the same reason. All the samples taken from the top tap was analyzed and compared to one another to figure out where the composition of the top samples was becoming approximately constant, thus figuring out when the steady state was reached.

3.4 Exercise nr. 2

In the second exercise the goal was to find out how the column efficiency varies as a function of the vapor velocity. The column was run with total reflux for three new and different boiler powers, 30%-, 50% and 60% power. After the column reached steady state for each of the power levels, which were assumed to be in the same time it reached steady state in exercise 1

3.3, 3-4 mL top-samples were drawn. The reflux rate where also noted at the sampling time. The samples were analyzed and the efficiency calculated

3.5 Column Shutdown

The heater was turned off and the content in the column was left to cool down. When it was cold enough the content was discarded, and the cooling water was turned off aswell.

3.6 Gas Chromatography Analysis

Samples for the GC was prepared by mixing 500 µL of ethanol sample with 250 µL of isopropanol in a glass vial. The vials were sealed properly with caps. Then analyzed by using gas chromatography.

4 Results

4.1 Exercise 1

All the samples from the first part of the experiment were analyzed for alcohol content, using gas chromatography. The molar fraction, x_{EtOH} was found using equation B.3. Table 4.1 contains the volume percentage of ethanol, vol$\%_{EtOH}$, and the molar fractions, x_{EtOH} for each sample in excercise 1.

Table 4.1: Reboiler (40%) Gass chromatography results.

Samples	Time [min]	vol [%]	x_{EtOH}
1	5	87.546	0.6852
2	10	89.585	0.7270
3	15	90.210	0.7405
4	20	91.470	0.7686
5	25	91.375	0.7664
6	30	91.429	0.7676
7	35	91.840	0.7770
8	40	91.928	0.7791
9	45	91.832	0.7769
10	50	91.436	0.7678
11	55	91.743	0.7748
12	60	91.743	0.7748

Figure 4.1 shows the molar fractions as a function of time. The samples were taken at a five-minute interval over the course of an hour at boiler duty 40% in order to determine when the column reached steady state.

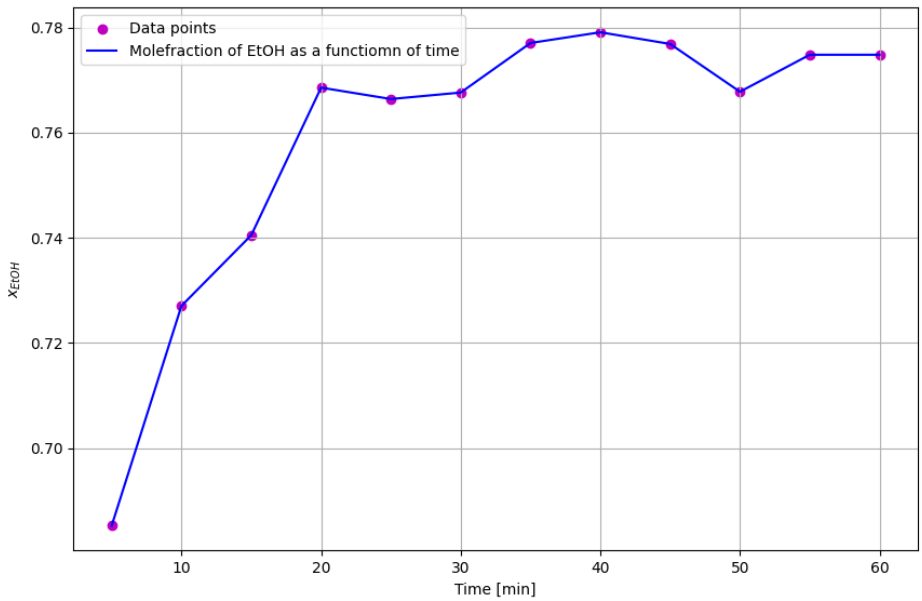

Figure 4.1: Molar fraction plotted as a function of time.

4.2 Exercise 2

Table 4.2 shows the volume fractions for distillate (D) and bottom product (B) at different power settings, as well as the flow and temperature in the distillation column.

Table 4.2: Gas chromatography results for different power settings for the reboiler.

Power setting [%]	Flow	Temperature	vol%$_D$	vol%x_B
30	30	75.4	87.55	16.08
40	39	74.8	91.93	17.65
50	58	74.5	93.14	19.53
60	65	74.5	90.94	17.41

Table 4.3 is an overview of the molar fractions of ethanol in the distillate and the bottom product. To find the molar fractions, te equation B.3 was used.

Table 4.3: The molar fraction of etanol in the destilate and the boiler for each rebioler power run.

Power setting [%]	x_D	x_B
30	0.6853	0.0560
40	0.7791	0.0622
50	0.8079	0.0699
60	0.7566	0.0613

4.2.1 McCabe-Thilele diagrams

McCabe-Thiele diagrams were plotted for each of the four experiments with different power settings. The diagrams are shown in the figures 4.2-4.5. The diagrams were plotted using data

given in D.2, and the operating line is $x = y$ because of total reflux. In the diagrams, x_D is the molar fraction of ethanol in the distillate, and x_B is the molar fraction of ethanol in the bottom product. The diagrams were used to find the ideal number of trays, which are shown in table 4.4.

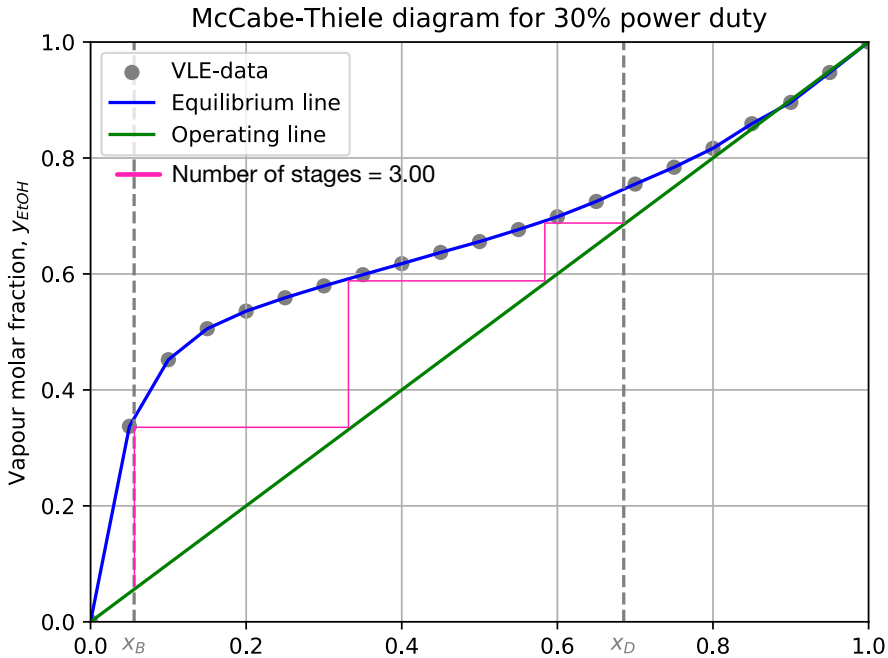

Figure 4.2: mcCabe-thiele diagram for reboiler power 30. The ideal number of trays is 3.

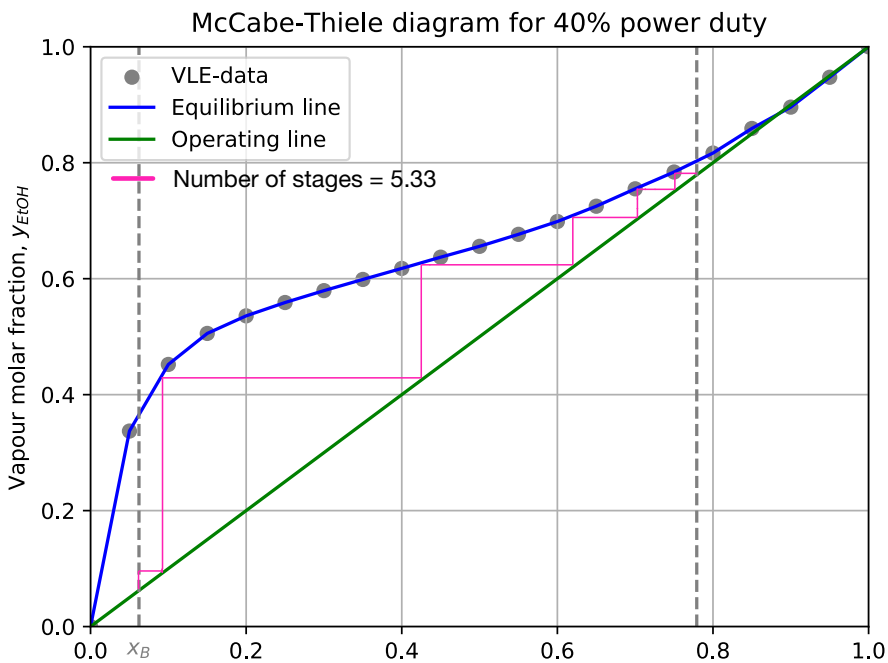

Figure 4.3: mcCabe-thiele diagram for reboiler power 40. The number of ideal trays is 5.33.

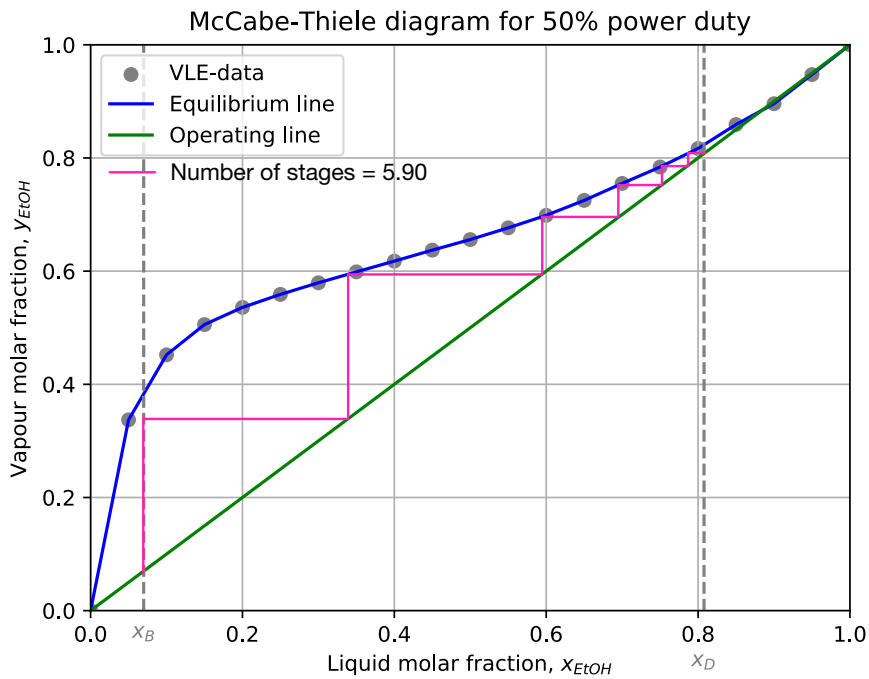

Figure 4.4: mcCabe-thiele diagram for reboiler power 50. The number of ideal trays is 5.90.

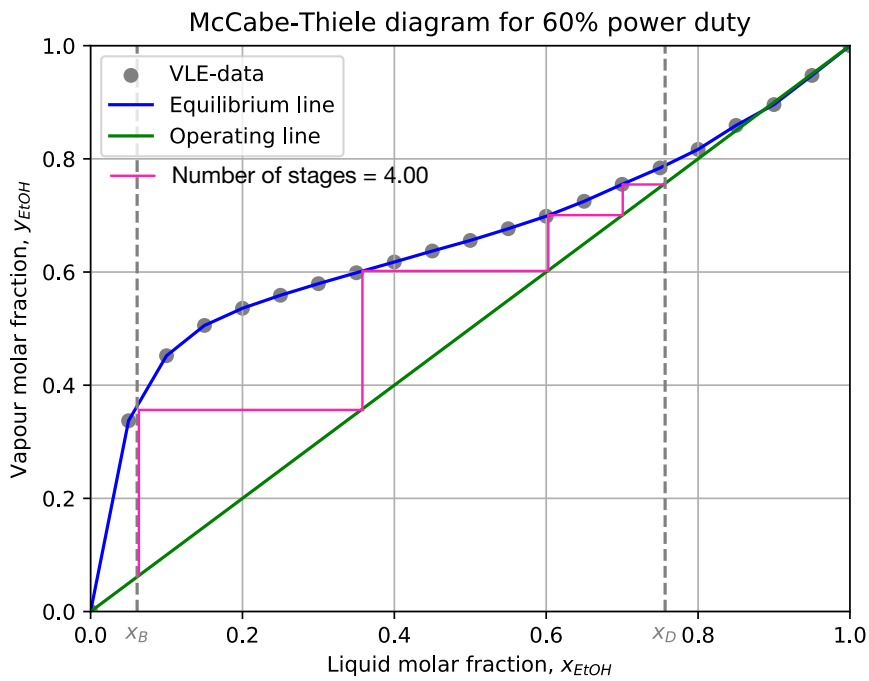

Figure 4.5: mcCabe-thiele diagram for reboiler power 60. The number of ideal trays is 4.

The theoretical stages were used to calculate Height Equivalent to a Theoretical Plate (HETP) using the equation B.4. The vapour velocity for each power setting was found using equation

B.8. HETP and vapour velocity can be found in table 4.4. Figure 4.6 shows the HETP for different runs plotted against vapor velocity. The labels on the plot indicate the reboiler power setting for each run.

Table 4.4: Theoretical stages, HETP and vapor velocity for each of the different reboiler power duties.

Power setting [%]	Theoretical stages	HETP [m]	Vapour velocity [ms^{-1}]
30	3.00	0.383	0.256
40	5.33	0.215	0.234
50	5.90	0.195	0.227
60	4.00	0.288	0.239

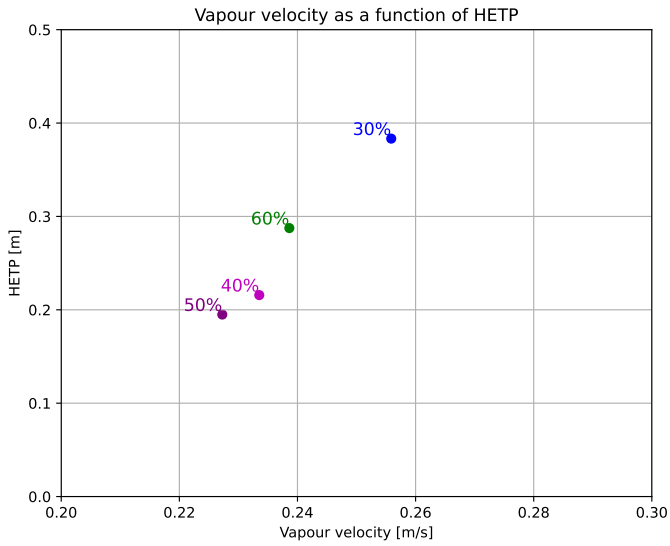

Figure 4.6: Plot of HETP and velocity.

5 Discussion

5.1 Time Required to Reach Steady State

In the first run, the percentage of ethanol in the top samples increased almost consistently until it began to decrease after 25 minutes. It then rose again until 45 minutes, at which point it started to decline once more. In the end, there was another increase, suggesting that it may have reached a steady state. However, this "steady state" does not represent the highest concentration of ethanol, which was 0.7791 at 40 minutes. This indicates that the data from this experiment cannot be used to determine the time required to reach steady state.

The total time for the runs in experiment two was based on the assumption that steady state is reached within forty minutes. Since this is not the case, it is not guaranteed that they achieved steady state either. The highest ethanol composition was reached during the first experiment after 40 minutes, while the second highest was observed in the top sample from the run with a reboiler power of 50. Therefore, neither the steady state time nor the steady state ethanol composition for this experiment is known.

Flooding was not observed during the first run; however, if it had occurred, it would have reduced the contact between the gas and the liquid. This contact is essential for cooling

the vapor, allowing water to condense and flow back to the bottom while the ethanol vapor rises. Reduced interaction between the liquid and gas phases makes the separation process less effective. Additionally, flooding could cause the reflux to flow back to the top sampling point rather than descending through the packed column.

Flooding can also result in an increase in pressure, which raises the boiling points of the components involved. This elevation in boiling point can lead to reduced evaporation of ethanol.

Unstable temperature could also influence the effect of the separation. A lower temperature may result in less effective evaporation of ethanol, while a higher temperature could cause increased evaporation of water. Both scenarios would lead to a less effective separation and a decrease in ethanol concentration.

If the solution was a heterogeneous mixture, which it likely is, the ethanol concentration would vary in different parts of the solution. This uneven distribution could lead to inconsistent evaporation rates, potentially affecting both the efficiency of the separation process and the final ethanol concentration.

5.2 Column Efficiency and Vapor Velocity

The lack of information on steady-state temperature and the time required to reach it impacts the results of experiment two. However, some insights can still be drawn from the data.

In theory, the reflux should increase with higher reboiler power and decrease with lower power. A higher reflux and reboiler power would result in a distillate with a higher ethanol content, indicating higher column efficiency. Since packed columns lack trays, HETP is used as an estimate. Higher column efficiency suggests better contact between liquid and vapor, achieved by a higher number of theoretical stages, which corresponds to a lower HETP.

Figure **??** shows a trend where higher reboiler power results in lower HETP, thus improving column efficiency. This trend, that aligns with the theory, is observed in the runs with reboiler powers ranging from 30% to 50%. However, at 60% reboiler power, the HETP increases, indicating reduced efficiency. This run a flooding occured. Flooding happens when the vapor pressure is too high for liquid to flow downward, leading to a distillate with lower ethanol concentration and decreased efficiency. The lower concentration is a result of the rerun returning to the top sampling point, instead of the boiler, and the vapour not being able to condense and flow ro the top sample point. These results suggest there is a maximum reboiler power, beyond which efficiency decreases due to flooding.

6 Conclusion

The experiments faced challenges in determining the time required to reach steady state. The highest ethanol concentration was observed after 40 minutes, but the data was inconsistent, suggesting that steady state was not definitively reached. Factors like potential flooding, unstable temperature, and heterogeneous mixing likely affected the separation efficiency. Vapor velocity increased with higher reboiler power, influencing the reflux rate and column efficiency. While the HETP data showed trends consistent with theory, the experiment should have started with the highest power level, because using the steady-state time for 40% power duty made it difficult to extract any data for the higher reboiler power duties. Additionally, excessive power such as 60% may have caused performance issues such as flooding and reduced efficiency

References

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A Calculations

A.1 A mixture of H_2O and EtOH

The physical properties needed when calculating the volumetric composition of the mix of water and ethanol, are shown in table A.1. Mm is molar mass, ρ is the density, and x is the component's mole fraction.

Table A.1: Physical properties for H_2O and EtOH

Component	$Mm[gmol^{-1}]$	$\rho[g/cm^3]$	x
EtOH	46.040	0.789	0.1
H_2O	18.016	0.997	0.9

The mixture's total volume, $V_{tot} = 5800mL$, can be expressed as:

$$V_{tot} = V_{\mathrm{H_2O}} + V_{EtOH}, \tag{A.1}$$

where $V_{\mathrm{H_2O}}$ is the volume of H_2O, and V_{EtOH} is the volume of the ethanol. The mole fraction for ethanol in the mixture is given by:

$$x_{EtOH} = \frac{V_{\mathrm{H_2O}}}{n_{EtOH} + n_{\mathrm{H_2O}}}, \tag{A.2}$$

where n is the number of moles of the respective components. The equations A.3 and A.4 are used to calculate the moles for water and ethanol respectively.

$$n_{\mathrm{H_2O}} = \frac{V_{\mathrm{H_2O}} \cdot \rho_{\mathrm{H_2O}}}{Mm_{\mathrm{H_2O}}}, \tag{A.3}$$

$$n_{\mathrm{EtOH}} = \frac{V_{\mathrm{EtOH}} \cdot \rho_{\mathrm{EtOH}}}{Mm_{EtOH}}, \tag{A.4}$$

where V is the components volume. Putting the equations A.1, A.3, and A.4 into A.2, to get the following equation:

$$X_{EtOH} = \frac{\frac{V_{\mathrm{EtOH}} \cdot \rho_{\mathrm{EtOH}}}{Mm_{EtOH}}}{\frac{V_{\mathrm{EtOH}} \cdot \rho_{\mathrm{EtOH}}}{Mm_{EtOH}} + \frac{(V_{tot} - V_{EtOH}) \cdot \rho_{\mathrm{H_2O}}}{Mm_{\mathrm{H_2O}}}} \tag{A.5}$$

By inserting the variables from table A.1, A.5 can be solved for V_{EtOH}, which gives:

$$V_{EtOH,100\%} = 1532.3mL \tag{A.6}$$

Since 96% ethanol is used in the experiment, the calculated value above needs to be divided by 0.96 in order to find the correct volume,

$$V_{EtOH,96\%} = 1596.1mL \tag{A.7}$$

With $V_{EtOH,96\%}$ known,. it's possible to find $V_{\mathrm{H_2O}}$ by using the equation A.1,

$$V_{\mathrm{H_2O}} = 4203.9mL \tag{A.8}$$

B Result Calculations

B.1 Conversion of Volumetric Fraction into Molar Fraction

It is desired to calculate the mole fractions of ethanol in both the distillate and bottom product. x_{EtOH}, the molar fraction of ethanol can be expressed as

$$x_{EtOH} = \frac{n_{EtOH}}{n_{tot}} = \frac{n_{EtOH}}{n_{EtOH} + n_{H_2O}}, \tag{B.1}$$

where n_{EtOH} and n_{H_2O} is the amount of substance in the solution for ethanol and water, respectively. In order to convert the volume, V_i, of component i, into moles, for the component, n_i, the following equation is used

$$n_i = \frac{n_{\rho_i V_i}}{Mm_i} = \frac{\rho_i x_{V,i} V_{tot}}{Mm_i}, \tag{B.2}$$

where ρ_i is the density, V_i is the volume, Mm_i is the molar mass, and $x_{V,i}$ is the volume fraction of the component i. V_{tot} is the total volume for the solution.

By inserting equation B.2 into B.1, the following equation is obtained

$$\begin{aligned} x_{EtOH} &= \frac{\frac{\rho_{EtOH} \cdot x_{V,EtOH} \cdot V_{tot}}{Mm_{EtOH}}}{\frac{\rho_{EtOH} \cdot x_{V,EtOH} \cdot V_{tot}}{Mm_{EtOH}} + \frac{\rho_{H_2O} \cdot x_{V,H_2O} \cdot V_{tot}}{Mm_{H_2O}}} \\ &= \frac{\frac{\rho_{EtOH} \cdot x_{V,EtOH}}{Mm_{EtOH}}}{\frac{\rho_{EtOH} \cdot x_{V,EtOH}}{Mm_{EtOH}} + \frac{\rho_{H_2O} \cdot x_{V,H_2O}}{Mm_{H_2O}}} \end{aligned} \tag{B.3}$$

B.2 Height Equivalent

The height equivalent to a theoretical plate (HETP) is calculated by using the following equation

$$HETP = \frac{H}{N_t}, \tag{B.4}$$

where H is the height of the distillation column, and N_t is the number of theoretical stages found in the McCabe-Thiele-diagrams. The height of the distillation column is given in D.1, and and the number of theoretical stages are shown in table 4.4.

B.3 Vapour Velocity

To calculate the vapour velocity, the following equation is used

$$v = \frac{\dot{V}_{gas}}{A}, \tag{B.5}$$

where $\dot{V}_{gas}$ is the volume rate of the gas, and A is the cross-sectional area of the Column. To calculate A, the following equation is used

$$A = \frac{\pi}{4} D^2, \tag{B.6}$$

where D is the column's diameter. The ideal gas law can be used to calculate the volume rate, $\dot{V}_{gas}$,

$$\dot{V}_{gas} = \frac{\dot{n}RT}{p}, \tag{B.7}$$

where $\dot{n}$ is the molar gas flow, R is the universal gas constant, T is the temperature, and p is the pressure. The vapour velocity can be calculated by putting the equations B.6 and B.7 into B.5

$$v = \frac{4\dot{n}RT}{\pi D^2 p}, \tag{B.8}$$

the D, T and p is given in the figure D.1. In order to calculate v, the molar flow needs to be found

$$\dot{n} = \frac{\dot{m}}{Mm} = \frac{\dot{V} \cdot \rho_{\text{distillate}}}{x_{\text{EtOH}} \cdot Mm_{\text{EtOH}} + (1 - x_{\text{EtOH}}) \cdot Mm_{\text{H2O}}} \tag{B.9}$$

where $\dot{V}$ is the total reflux, and x_{EtOH} is the molar fraction of ethanol. These values are given in the table 4.1. $\rho_{distillate}$ was calculated using the equation

$$\rho_{distillate} = \rho_{EtOH} \cdot x_{EtOH} + \rho_{H_2O} \cdot x_{H_2O} \tag{B.10}$$

The $\rho_{distillate}$-value from equation B.10 is put into B.9. When $\dot{n}$ is found, it can be used in the equation B.8 to find the vapour velocity.

C Hand drawn McCabe-Thiele

Figure C.1 shows a hand-drawn McCabe-Thiele diagram for the run with a reboiler power of 30. The diagram shows that the theoretical number of stages is approximately 3.5.

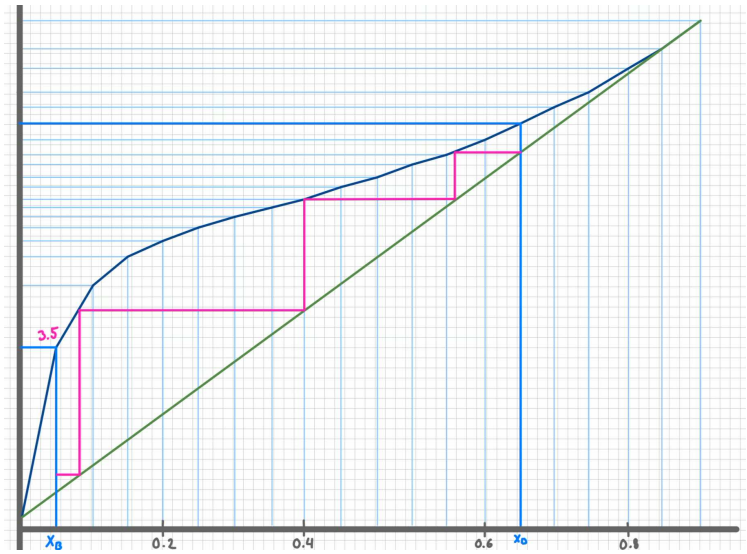

Figure C.1: Handdrawn McCabe-Thiele diagram.

D Provided data

Both known variables and constants and equilibrium data was provided and are listed below

D	0.05 m
H	1.15 m
V_{total}	5800 mL
$\dot{V}$	53 mL/min
T	298 K
p	10^5 Pa
$Mm_{ethanol}$	46.1 g/mol
Mm_{water}	18.0 g/mol
$\rho_{ethanol}$	0.785 g/cm^3
ρ_{water}	1.0 g/cm^3
R	8.314 PaL/molK

Figure D.1: Known variables and constants.

x_1	y_1
0	0
0.05	0.3372
0.100	0.4521
0.15	0.5056
0.200	0.5359
0.250	0.5589
0.300	0.5794
0.350	0.5987
0.400	0.6177
0.450	0.6371
0.500	0.6558
0.550	0.6765
0.600	0.6986
0.650	0.7250
0.700	0.7550
0.750	0.7840
0.800	0.8167
0.850	0.8591
0.900	0.8959
0.950	0.9474
1	1

Figure D.2: Equilibrium data for Ethanol water system.

E Python code

```
import numpy as np
import matplotlib.pyplot as plt

#konstanter
roe_EtOH = 0.789
roe_H2O = 0.997
Mm_EtOH = 46.040
Mm_H2O = 18.016

t = np.array(np.linspace(5, 60, 12))
V_prosent = np.array([87.546, 89.585, 90.210, 91.470, 91.375, 91.429, 91.840, 91.928, 91.832, 91.436, 91.74
V_pr_H2O = 100 - V_prosent

# Molfraksjonen til etanol regnes med B4
def B4(i, func_EtOH, func_H2O):
    x = ((roe_EtOH*func_EtOH[i]) / Mm_EtOH) / (((roe_EtOH*func_EtOH[i]) / Mm_EtOH) + ((roe_H2O * func_H2O[i
    return float(x)

# Bruker B4 og putter alle verdiene i en liste x_EtOH for Excercise 1
def xetoh(func_EtOH, func_H2O):
    x_EtOH = []
    for i in range(len(func_EtOH)):
        x_EtOH.append(B4(i, func_EtOH, func_H2O))
    return x_EtOH

x_etoh = xetoh(V_prosent, V_pr_H2O)

plt.scatter(t, x_etoh, color = 'm', label = 'Data points')
plt.plot(t, x_etoh, color = 'b', label = 'Molefraction of EtOH as a functiomn of time')
plt.xlabel('Time [min]')
plt.ylabel(r'$x_{EtOH}$')
plt.legend()
plt.grid(True)
plt.show()


#Excercise 2

reboiler = np.array([30, 40, 50, 60])
flow = np.array([30, 39, 58, 65])
T = np.array([75.4, 74.8, 74.5, 74.5])
xv_D = np.array([0.8755, 0.9193, 0.9314, 0.9094]) * 100
xv_D_h2o = 100 - xv_D
xv_B = np.array([0.1608, 0.1765, 0.1953, 0.1741]) * 100
xv_B_h2o = 100 - xv_B

# molfraksjoner av EtOH for destillat og bunnprodukt. Listen gir verdiene for pumpenivå 30, 40, 50, 60
x_D = np.array(xetoh(xv_D, xv_D_h2o))
x_B = np.array(xetoh(xv_B, xv_B_h2o))

#equilibrium data for etanol i vann
x1 = [0, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55,
      0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 1]
y1 = [0, 0.3372, 0.4521, 0.5056, 0.5359, 0.5589, 0.5794, 0.5987, 0.6177,
      0.6371, 0.6558, 0.6765, 0.6986, 0.725, 0.755, 0.784, 0.8167, 0.8591,
      0.8959, 0.9474, 1]
```

```
#operating line er lineær og går fra origo til (1,1), lager to minilister for å gjøre plottet enkelt

plt.scatter(x1, y1, color = 'grey', label = 'VLE-data')
plt.plot(x1, y1, color = 'b', label = 'Equilibrium line')
plt.plot([0, 1], [0, 1], color = 'green', label = 'Operating line')
plt.axvline(x=x_D[0], color='grey', linestyle='--')
plt.axvline(x=x_B[0], color='grey', linestyle='--')
plt.text(x_D[0], -0.05, r'$x_D$', color='grey', ha='center')
plt.text(x_B[0], -0.05, r'$x_B$', color='grey', ha='center')
plt.xlim(0, 1)
plt.ylim(0, 1)
plt.title(r'McCabe-Thiele diagram for 30$\%$ power duty')
plt.xlabel(r'Liquid molar fraction, $x_{EtOH}$')
plt.ylabel(r'Vapour molar fraction, $y_{EtOH}$')
plt.legend(loc='upper left')
plt.grid(True)
plt.show()

# Verdier fra McCabe-Thiele-grafene
Nt = np.array([3.00, 5.33, 5.90, 4.00])

def func_HETP():
    HETP = []
    H = 1.15
    for i in range(len(Nt)):
        HETP.append(float(H / Nt[i]))
    return HETP

# Vapour velocity!
def vap_vel(n_dot):
    R = 8.314
    T = 298
    D = 0.05
    p = 10**5
    v = []
    for i in range(len(n_dot)):
        v.append((4*n_dot[i]*R*T)/(np.pi*D**2 * p))
    return v

# Må beregne tettheten til løsningen
def roe_solution(i):
    roe_frac_EtOH = x_D[i] * roe_EtOH
    roe_frac_H2O = (1 - x_D[i]) * roe_H2O
    roe_D = roe_frac_EtOH + roe_frac_H2O
    return roe_D

# Beregner molstrøm
def n_flow():
    n_dot = []
    V_flow = 53/60       # mL/s
    for i in range(len(x_D)):
        n_dot.append(float((V_flow * roe_solution(i)) / (x_D[i] * Mm_EtOH + (1 - x_D[i]) * Mm_H2O)))
    return n_dot

n_dot = n_flow()
v_vap = vap_vel(n_dot)
HETP = func_HETP()
# Want to label the data-points I'll get when plotting
labels = ['30%', '40%', '50%', '60%']
```

```
# Want different colours for the points as well
colours = ['b', 'm', 'purple', 'green']



plt.scatter(v_vap, HETP, color = colours)
for i in range(len(v_vap)):
    plt.text(v_vap[i], HETP[i], labels[i], fontsize=12, ha='right', va='bottom', color=colours[i])
plt.title('Vapour velocity as a function of HETP')
plt.xlabel('Vapour velocity [m/s]')
plt.ylabel('HETP [m]')
plt.grid(True)
plt.xlim(0.2, 0.3)
plt.ylim(0, 0.5)
plt.show()
```